

Zimo (Zina) Ma

Chicago, IL (Open to Relocate) | +1 (765) 712-2173 | zimoma0819@gmail.com | [LinkedIn](#)

SUMMARY

- **Self:** An aspiring Data Scientist with 4+ years academic data analytics and research experience, with strong communication skills, passionate about building machine learning models for data-driven strategic decision-making.
- **Skills:** Data Analytics (Data Mining, Data Visualization, Metadata Management), Machine Learning (Regression, Decision Tree, Random Forest, K-Means, XGBoost, LightGBM, CatBoost), Statistics (Probability, Hypothesis Testing, PCA), Data Structures and Algorithms in Python, Web Scraping, Marketing Strategy, A/B Testing, Microeconomics, Project Management
- **Technical:** Python (NumPy, Pandas, SciPy, Seaborn, Matplotlib, Statsmodels, Scikit-learn), SQL, Java, R, GitHub, Shiny App, Tableau, Power BI, SPSS, AWS, Excel (Pivot Table, VLOOKUP, VBA), Qualtrics; Google Advanced Data Analytics Professional Certificate

EDUCATION

University of Chicago | Chicago, IL

Aug 2023 - Jun 2025

Master of Arts in Computational Social Science

GPA: 3.88/4.0

DePauw University | Greencastle, IN

Jul 2019 - May 2023

Bachelor of Arts in Mathematics and Psychology

GPA: 3.72/4.0

PROFESSIONAL EXPERIENCE

Data Engineer | Huatai Life Insurance Company, Beijing CN

Jun 2024 - Aug 2024

- Upgraded company-wide policy search engine by optimizing the sorting logic of the policy list and implementing real-time data filters in **Python**, resulting in a 75% improvement in efficiency and faster data-driven insights for policy analysis
- Tested and refactored legacy codes for insurance policy database over 4,000 files based on Sonar system scan results in **Python** to ensure the integration and unified management of all policy data for convenient access
- Led a team of 4 interns and analyzed portfolio health and future profitability, conducted cross-team collaboration with financial department, marketing strategy and proposal adopted by senior leadership and stakeholders

Research Analyst | DePauw University, Greencastle IN

The Environment is Sexist: Is Women's Oppression Linked with Nature? (advised by Dr. Naima Shifa)

Aug 2022 - May 2023

- Conducted data visualization, **hypothesis testing & correlation tests** on Demographic and Health Survey data in **GIS** and **R** to analyze the association between Indian women's disproportional death against natural disasters and socioeconomic factors
- Presented the findings at the Challenging Borders in Domestic & International Human Rights Conference funded by the Great Lakes Colleges Association in front of 100+ students representing 18 higher-ed schools from 10 countries

COVID-19 is Not All Bad News: Implications for STEM Instruction (advised by Dr. Pamela Propsom)

Mar 2021 - May 2023

- Performed **Chi-Square tests** and independent **t-tests** using **SPSS** over 391-participant survey data to identify relations between COVID-19 and STEM learning across diverse student groups such as underrepresented groups
- Identified domestic students reported more negative impact of COVID on their STEM experience than international students and other minorities, and presented findings to all 50 STEM faculty members to improve 12 STEM Guide courses quality
- Submitted the journal article under the review of *The Journal of College Science Teaching* and presented the applied social psychology poster to the 41st Annual Mid-America Undergraduate Psychology Research Conference

RESEARCH & PROJECTS

TikTok User Claims/Opinions Classification | **Data Analyst**

Mar 2024 - Oct 2024

- Constructed an end-to-end AI-based solution to determine whether a TikTok video contains a claim or offers an opinion, to help reduce the backlog of user reports and allow moderators to focus on content requiring immediate attention
- Performed **hypothesis testing** and a **2-sample t-test** between verified vs unverified users, showing average views for videos posted by verified accounts are significantly lower than unverified accounts and delivering an executive summary via **Tableau**
- Implemented a **Logistic Regression** model in **Python** to predict TikTok user verification status, achieving 90% recall for verified users by effectively managing data imbalances (**SMOTE**) and refining video engagement metrics
- Built and optimized **Random Forest** and **XGBoost** classifier to determine the nature of TikTok videos as claims or opinions, fine-tuned model hyper-parameters, achieving classification accuracy of 99.61% and 99.55% respectively

Machine Learning-Driven MBTI Classification | **Machine Learning Engineer**

Jan 2024 - Mar 2024

- Pre-processed (data cleaning, lemmatization, and normalization) 8k people's MBTI personality types (16-classification) and their recent 50+ social media posts using tf-idf vectorizer, and conducted feature engineering with **PCA**
- Developed **Logistic Regression**, **Random Forest** and **Truncated SVD** in **Python** for MBTI prediction, improved accuracy by 24% and identified extroverted (E) & preceptive (P) personality types are prone to reveal personality traits via social media

Course Search Engine Backend Development | **Backend Developer**

Jan 2024 - Feb 2024

- Implemented and integrated 30+ complex **SQL** queries using **SQLite3** in **Python** to integrate course catalog and schedule of 2,400 UChicago courses into an advanced filtering system, improving system efficiency by 42%
- Designed the **Django**-based web interface on the front end to create more user-friendly interactions for course selection, conducted unit testing and integral (end-to-end) testing before deployment